KU LEUVEN

FACULTY OF
ENGINEERING SCIENCE

The best master's thesis ever

First Author
Second Author

Thesis submitted for the degree of
Master of Science in
Electrical Engineering, option
Electronics and Chip Design

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Preface

This is my acknowledgment to thank everyone who has kept me busy. I would like to thank my supervisor, my advisor, and the entire jury. My family has also been very supportive, of course.

First Author
Second Author

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Abstract

This `abstract` environment provides a summary of the work, which may or may not be extensive. The intention is that this should be limited to 1 page.

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

List of Figures and Tables

List of Figures

3.1 The logo of the Faculty of Engineering Science. 6

List of Tables

3.1 A table with the wrong layout. 6
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List of abbreviations and symbols

abbreviations

LoG	Laplacian-of-Gaussian
MSE	Mean Square error
PSNR	Peak Signal-to-Noise ratio

Symbols

42	"The Answer to the Ultimate Question of Life, the Universe, and Everything" volgens de [1]
c	Speed of light
E	Energy
m	Mass
π	The number pi

Chapter 1

Introduction

The first contains a general introduction to the work. The goals are defined and the modus operandi is explained.

1.1 Lorem Ipsum 4–5

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetuer.

1.2 Lorem Ipsum 6–7

Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.

Sed commodo posuere pede. Mauris ut est. Ut quis purus. Sed ac odio. Sed vehicula hendrerit sem. Duis non odio. Morbi ut dui. Sed accumsan risus eget odio. In hac habitasse platea dictumst. Pellentesque non elit. Fusce sed justo eu urna porta tincidunt. Mauris felis odio, sollicitudin sed, volutpat a, ornare ac, erat. Morbi quis dolor. Donec pellentesque, erat ac sagittis semper, nunc dui lobortis purus, quis congue purus metus ultricies tellus. Proin et quam. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent sapien turpis, fermentum vel, eleifend faucibus, vehicula eu, lacus.

Chapter 2

The First Chapter

A chapter is a logical unit. It normally starts with an introduction, which you are reading now. The last topic of the chapter holds the conclusion.

2.1 The First Topic of the Chapter

First comes the introduction to this topic.

Nunc velit. Nullam elit sapien, eleifend eu, commodo nec, semper sit amet, elit. Nulla lectus risus, condimentum ut, laoreet eget, viverra nec, odio. Proin lobortis. Curabitur dictum arcu vel wisi. Cras id nulla venenatis tortor congue ultrices. Pellentesque eget pede. Sed eleifend sagittis elit. Nam sed tellus sit amet lectus ullamcorper tristique. Mauris enim sem, tristique eu, accumsan at, scelerisque vulputate, neque. Quisque lacus. Donec et ipsum sit amet elit nonummy aliquet. Sed viverra nisl at sem. Nam diam. Mauris ut dolor. Curabitur ornare tortor cursus velit.

2.1.1 An item

Please don't abuse enumerations: short enumerations shouldn't use "`itemize`" or "`enumerate`" environments. So *never write*:

> The Eiffel tower has three floors:
>
> - the first one;
> - the second one;
> - the third one.

But write:

> The Eiffel tower has three floors: the first one, the second one, and the third one.

2.2 A Second Topic

First we start with equation 2.1 after this paragraph.

$$A = \pi r^2 \tag{2.1}$$

Vivamus sit amet pede. Duis interdum, nunc eget rutrum dignissim, nisl diam luctus leo, et tincidunt velit nisl id tellus. In lorem tellus, aliquet vitae, porta in, aliquet sed, lectus. Phasellus sodales. Ut varius scelerisque erat. In vel nibh eu eros imperdiet rutrum. Donec ac odio nec neque vulputate suscipit. Nam nec magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Nullam porta, odio et sagittis iaculis, wisi neque fringilla sapien, vel commodo lorem lorem id elit. Ut sem lectus, scelerisque eget, placerat et, tincidunt scelerisque, ligula. Pellentesque non orci.

2.2.1 Another item

Morbi tincidunt posuere arcu. Cras venenatis est vitae dolor. Vivamus scelerisque semper mi. Donec ipsum arcu, consequat scelerisque, viverra id, dictum at, metus. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut pede sem, tempus ut, porttitor bibendum, molestie eu, elit. Suspendisse potenti. Sed id lectus sit amet purus faucibus vehicula. Praesent sed sem non dui pharetra interdum. Nam viverra ultrices magna.

Aenean laoreet aliquam orci. Nunc interdum elementum urna. Quisque erat. Nullam tempor neque. Maecenas velit nibh, scelerisque a, consequat ut, viverra in, enim. Duis magna. Donec odio neque, tristique et, tincidunt eu, rhoncus ac, nunc. Mauris malesuada malesuada elit. Etiam lacus mauris, pretium vel, blandit in, ultricies id, libero. Phasellus bibendum erat ut diam. In congue imperdiet lectus.

2.3 Conclusion

The final section of the chapter gives an overview of the important results of this chapter. This implies that the introductory chapter and the concluding chapter don't need a conclusion.

Nunc sed pede. Praesent vitae lectus. Praesent neque justo, vehicula eget, interdum id, facilisis et, nibh. Phasellus at purus et libero lacinia dictum. Fusce aliquet. Nulla eu ante placerat leo semper dictum. Mauris metus. Curabitur lobortis. Curabitur sollicitudin hendrerit nunc. Donec ultrices lacus id ipsum.

Chapter 3

The Next Chapter

Vivamus adipiscing. Curabitur imperdiet tempus turpis. Vivamus sapien dolor, congue venenatis, euismod eget, porta rhoncus, magna. Proin condimentum pretium enim. Fusce fringilla, libero et venenatis facilisis, eros enim cursus arcu, vitae facilisis odio augue vitae orci. Aliquam varius nibh ut odio. Sed condimentum condimentum nunc. Pellentesque eget massa. Pellentesque quis mauris. Donec ut ligula ac pede pulvinar lobortis. Pellentesque euismod. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent elit. Ut laoreet ornare est. Phasellus gravida vulputate nulla. Donec sit amet arcu ut sem tempor malesuada. Praesent hendrerit augue in urna. Proin enim ante, ornare vel, consequat ut, blandit in, justo. Donec felis elit, dignissim sed, sagittis ut, ullamcorper a, nulla. Aenean pharetra vulputate odio.

3.1 The First Topic of this Chapter

Quisque enim. Proin velit neque, tristique eu, eleifend eget, vestibulum nec, lacus. Vivamus odio. Duis odio urna, vehicula in, elementum aliquam, aliquet laoreet, tellus. Sed velit. Sed vel mi ac elit aliquet interdum. Etiam sapien neque, convallis et, aliquet vel, auctor non, arcu. Aliquam suscipit aliquam lectus. Proin tincidunt magna sed wisi. Integer blandit lacus ut lorem. Sed luctus justo sed enim.

3.1.1 An item

A master's thesis is never an isolated work. This means that your text must contain references. On-line documents[3] as well as books[2] can be referenced.

Usually a text also contains inline formulas ($\sin^2\eta + \cos^2\eta = 1$) or formulas as separate equations.

$$\sigma(t) = \frac{1}{\sqrt{2\pi}} \int_0^t e^{-x^2/2} dx \tag{3.1}$$

Figure 3.1: The logo of the Faculty of Engineering Science.

gnats	gram	$13.65
	each	.01
gnu	stuffed	92.50
emu		33.33
armadillo	frozen	8.99

Table 3.1: A table with the wrong layout.

Item		
Animal	Description	Price ($)
Gnat	per gram	13.65
	each	0.01
Gnu	stuffed	92.50
Emu	stuffed	33.33
Armadillo	frozen	8.99

Table 3.2: A table with the correct layout.

3.2 Figures

Figures are used to add illustrations to the text. The Figure 3.1 shows the KU Leuven logo as an illustration.

3.3 Tables

Tables are used to present data neatly arranged. A table is normally not a spreadsheet! Compare Table 3.1 en Table 3.2: which table do you prefer?

3.4 Lorem Ipsum

This section is added to check headers and footers. So this chapter must at least contain three pages. To make sure that we get the required amount, the lipsum

package isn't used but the text is put directly in the text.

3.4.1 Lorem ipsum dolor sit amet, consectetur adipiscing elit

Sed nec tortor id felis tristique sodales. Nulla nec massa eu dui fermentum tincidunt. Integer ullamcorper ante eget eros posuere faucibus. Nam id ligula ut augue pulvinar vulputate id at purus. Aenean condimentum tortor eu mi placerat eget eleifend massa mollis. Nam est mi, sagittis quis euismod eget, sagittis in nibh. Proin elit turpis, aliquam et imperdiet sed, volutpat eu turpis.

Pellentesque vel enim tellus, vitae egestas turpis. Praesent malesuada elit non nisi sollicitudin non blandit lacus tincidunt. Morbi blandit urna at lectus ornare laoreet. Suspendisse turpis diam, lobortis dictum luctus quis, commodo at lorem. Integer lacinia convallis ultricies. Sed quis augue neque, eu malesuada arcu. Nullam vehicula, purus vitae sagittis pulvinar, erat eros semper massa, eu egestas nibh erat quis magna. Cras pellentesque, nisl eu dapibus volutpat, urna augue ornare quam, quis egestas lectus nulla a lectus.

Vivamus dictum libero in massa cursus sed vulputate eros imperdiet. Donec lacinia, libero ac lobortis egestas, nibh dui ornare arcu, luctus porttitor velit massa sit amet quam. Maecenas scelerisque laoreet diam, vitae congue quam adipiscing vitae. Aliquam cursus nisl a leo convallis eleifend fermentum massa porta. Nunc libero quam, dapibus dapibus molestie sit amet, faucibus vel nunc.

3.4.2 Praesent auctor venenatis posuere

Sed tellus augue, molestie in pulvinar lacinia, dapibus non ipsum. Fusce vitae mi vitae enim ullamcorper hendrerit eu malesuada est. Proin iaculis ante sed nibh tincidunt vel interdum libero posuere. Vivamus accumsan metus quis felis congue suscipit dapibus enim mattis. Fusce mattis tortor eget ipsum interdum sagittis auctor id metus.

Integer diam lacus, pharetra sit amet tempor et, tristique non lorem. Aenean auctor, nisi eu interdum fermentum, lectus massa adipiscing elit, sed facilisis orci odio a lectus. Proin mi nibh, tempus quis porta a, viverra quis enim. In sollicitudin egestas libero, quis viverra velit molestie eget. Nulla rhoncus, dolor a mollis vestibulum, lacus elit semper nisi, nec sollicitudin sem urna eu magna. Nunc sed est urna, euismod congue mi.

3.4.3 Cras vulputate ultricies venenatis

Vivamus eros urna, sodales accumsan semper vel, lobortis sit amet mauris. Etiam condimentum eleifend lorem, ullamcorper ornare lectus aliquet vitae. Praesent massa enim, interdum sit amet semper et, venenatis ut elit. Quisque faucibus, quam ac lacinia imperdiet, nulla neque elementum purus, tempus rutrum justo massa porta sapien. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Sed ultrices interdum mi, et rhoncus sapien rutrum sed.

Duis elit orci, molestie quis sollicitudin sed, convallis non ante. Maecenas tincidunt condimentum justo, et ultricies leo tristique vitae. Vestibulum quis quam non lectus dapibus eleifend a vitae nibh. Nam nibh justo, pharetra quis iaculis consequat, elementum quis justo. Etiam mollis lacinia lacus, nec sollicitudin urna lobortis ac. Nulla facilisi.

Proin placerat risus eleifend erat ultricies placerat. Etiam rutrum magna nec turpis euismod consectetur. Phasellus tortor odio, lacinia imperdiet condimentum sed, faucibus commodo erat. Phasellus sed felis id ante placerat ultrices. Aenean tempor justo in tortor volutpat eu auctor dolor mollis. Aenean sit amet risus urna. Morbi viverra vehicula cursus.

3.4.4 Donec nibh ante, consectetur et posuere id, tempus nec arcu

Curabitur a tellus aliquet ipsum pellentesque scelerisque. Etiam congue, risus et volutpat rutrum, est purus dapibus leo, non cursus metus felis eget ligula. Vivamus facilisis tristique turpis, ut pretium lectus luctus eleifend. Fusce magna sapien, ullamcorper vitae fringilla id, euismod quis ante.

Phasellus volutpat, nunc et pharetra semper, sem justo adipiscing mauris, id blandit magna quam et orci. Vestibulum a erat purus, ut molestie ante. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Proin turpis diam, consequat ut ullamcorper ut, consequat eu orci. Sed metus risus, fringilla nec interdum vel, interdum eu nunc. Suspendisse vel sapien orci.

3.4.5 Morbi et mauris tempus purus ornare vehicula

Mauris sit amet diam quam, eget luctus purus. Sed faucibus, risus semper eleifend iaculis, mi turpis bibendum nisl, quis cursus nibh nisl sit amet ipsum. Vestibulum tempor urna vitae mi auctor malesuada eget non ligula. Nullam convallis, diam vel ultrices auctor, eros eros egestas elit, sed accumsan arcu tortor eget leo. Vestibulum orci purus, porttitor in pharetra eget, tincidunt eget nisl. Nullam sit amet nulla dui, facilisis vestibulum dui.

Donec faucibus facilisis mauris ac cursus. Duis rhoncus quam sed nisi laoreet eu scelerisque massa tincidunt. Vivamus sit amet libero nec arcu imperdiet tempor quis non libero. Sed consequat dignissim justo. Phasellus ullamcorper, velit quis posuere vulputate, felis erat tincidunt mauris, at vestibulum justo lectus et turpis. Maecenas lacinia convallis euismod. Quisque egestas fermentum sapien eu dictum. Sed nec lacus in purus dictum consequat quis vel nisl. Fusce non urna sem. Curabitur eu diam vitae elit accumsan blandit. Nullam fermentum nunc et leo dictum laoreet. Donec semper varius velit vel fringilla. Vivamus eu orci nunc.

3.5 Conclusion

The final section of the chapter gives an overview of the important results of this chapter. This implies that the introductory chapter and the concluding chapter don't need a conclusion.

Nunc sed pede. Praesent vitae lectus. Praesent neque justo, vehicula eget, interdum id, facilisis et, nibh. Phasellus at purus et libero lacinia dictum. Fusce aliquet. Nulla eu ante placerat leo semper dictum. Mauris metus. Curabitur lobortis. Curabitur sollicitudin hendrerit nunc. Donec ultrices lacus id ipsum.

Chapter 4

The Final Chapter

Morbi malesuada hendrerit dui. Nunc mauris leo, dapibus sit amet, vestibulum et, commodo id, est. Pellentesque purus. Pellentesque tristique, nunc ac pulvinar adipiscing, justo eros consequat lectus, sit amet posuere lectus neque vel augue. Cras consectetuer libero ac eros. Ut eget massa. Fusce sit amet enim eleifend sem dictum auctor. In eget risus luctus wisi convallis pulvinar. Vivamus sapien risus, tempor in, viverra in, aliquet pellentesque, eros. Aliquam euismod libero a sem.

4.1 The First Topic of this Chapter

4.1.1 Item 1

Sub-item 1

Nunc velit augue, scelerisque dignissim, lobortis et, aliquam in, risus. In eu eros. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Curabitur vulputate elit viverra augue. Mauris fringilla, tortor sit amet malesuada mollis, sapien mi dapibus odio, ac imperdiet ligula enim eget nisl. Quisque vitae pede a pede aliquet suscipit. Phasellus tellus pede, viverra vestibulum, gravida id, laoreet in, justo. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Integer commodo luctus lectus. Mauris justo. Duis varius eros. Sed quam. Cras lacus eros, rutrum eget, varius quis, convallis iaculis, velit. Mauris imperdiet, metus at tristique venenatis, purus neque pellentesque mauris, a ultrices elit lacus nec tortor. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent malesuada. Nam lacus lectus, auctor sit amet, malesuada vel, elementum eget, metus. Duis neque pede, facilisis eget, egestas elementum, nonummy id, neque.

Sub-item 2

Proin non sem. Donec nec erat. Proin libero. Aliquam viverra arcu. Donec vitae purus. Donec felis mi, semper id, scelerisque porta, sollicitudin sed, turpis. Nulla in urna. Integer varius wisi non elit. Etiam nec sem. Mauris consequat, risus nec

congue condimentum, ligula ligula suscipit urna, vitae porta odio erat quis sapien. Proin luctus leo id erat. Etiam massa metus, accumsan pellentesque, sagittis sit amet, venenatis nec, mauris. Praesent urna eros, ornare nec, vulputate eget, cursus sed, justo. Phasellus nec lorem. Nullam ligula ligula, mollis sit amet, faucibus vel, eleifend ac, dui. Aliquam erat volutpat.

4.1.2 Item 2

Fusce vehicula, tortor et gravida porttitor, metus nibh congue lorem, ut tempus purus mauris a pede. Integer tincidunt orci sit amet turpis. Aenean a metus. Aliquam vestibulum lobortis felis. Donec gravida. Sed sed urna. Mauris et orci. Integer ultrices feugiat ligula. Sed dignissim nibh a massa. Donec orci dui, tempor sed, tincidunt nonummy, viverra sit amet, turpis. Quisque lobortis. Proin venenatis tortor nec wisi. Vestibulum placerat. In hac habitasse platea dictumst. Aliquam porta mi quis risus. Donec sagittis luctus diam. Nam ipsum elit, imperdiet vitae, faucibus nec, fringilla eget, leo. Etiam quis dolor in sapien porttitor imperdiet.

4.2 The Second Topic

Cras pretium. Nulla malesuada ipsum ut libero. Suspendisse gravida hendrerit tellus. Maecenas quis lacus. Morbi fringilla. Vestibulum odio turpis, tempor vitae, scelerisque a, dictum non, massa. Praesent erat felis, porta sit amet, condimentum sit amet, placerat et, turpis. Praesent placerat lacus a enim. Vestibulum non eros. Ut congue. Donec tristique varius tortor. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Nam dictum dictum urna.

Phasellus vestibulum orci vel mauris. Fusce quam leo, adipiscing ac, pulvinar eget, molestie sit amet, erat. Sed diam. Suspendisse eros leo, tempus eget, dapibus sit amet, tempus eu, arcu. Vestibulum wisi metus, dapibus vel, luctus sit amet, condimentum quis, leo. Suspendisse molestie. Duis in ante. Ut sodales sem sit amet mauris. Suspendisse ornare pretium orci. Fusce tristique enim eget mi. Vestibulum eros elit, gravida ac, pharetra sed, lobortis in, massa. Proin at dolor. Duis accumsan accumsan pede. Nullam blandit elit in magna lacinia hendrerit. Ut nonummy luctus eros. Fusce eget tortor.

Ut sit amet magna. Cras a ligula eu urna dignissim viverra. Nullam tempor leo porta ipsum. Praesent purus. Nullam consequat. Mauris dictum sagittis dui. Vestibulum sollicitudin consectetuer wisi. In sit amet diam. Nullam malesuada pharetra risus. Proin lacus arcu, eleifend sed, vehicula at, congue sit amet, sem. Sed sagittis pede a nisl. Sed tincidunt odio a pede. Sed dui. Nam eu enim. Aliquam sagittis lacus eget libero. Pellentesque diam sem, sagittis molestie, tristique et, fermentum ornare, nibh. Nulla et tellus non felis imperdiet mattis. Aliquam erat volutpat.

4.3 Conclusion

Vestibulum sodales ipsum id augue. Integer ipsum pede, convallis sit amet, tristique vitae, tempor ut, nunc. Nam non ligula non lorem convallis hendrerit. Maecenas hendrerit. Sed magna odio, aliquam imperdiet, porta ac, aliquet eget, mi. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Vestibulum nisl sem, dignissim vel, euismod quis, egestas ut, orci. Nunc vitae risus vel metus euismod laoreet. Cras sit amet neque a turpis lobortis auctor. Sed aliquam sem ac elit. Cras velit lectus, facilisis id, dictum sed, porta rutrum, nisl. Nam hendrerit ipsum sed augue. Nullam scelerisque hendrerit wisi. Vivamus egestas arcu sed purus. Ut ornare lectus sed eros. Suspendisse potenti. Mauris sollicitudin pede vel velit. In hac habitasse platea dictumst.

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Chapter 5

Conclusion

The final chapter contains the overall conclusion. It also contains suggestions for future work and industrial applications.

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Appendices

Appendix A

The First Appendix

Appendices hold useful data which is not essential to understand the work done in the master's thesis. An example is a (program) source. An appendix can also have sections as well as figures and references[1].

A.1 More Lorem

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Appendix B

The Last Appendix

Appendices are numbered with letters, but the sections and subsections use arabic numerals, as can be seen below.

B.1 Lorem 20-24

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[1] D. Adams. *The Hitchhiker's Guide to the Galaxy.* Del Rey (reprint), 1995. ISBN-13: 978-0345391803.

[2] T. Pratchett and N. Gaiman. *Good Omens:* The Nice and Accurate Prophecies of Agnes Nutter, Witch. HarperTorch (reprint), 2006. ISBN-13: 978-0060853983.

[3] Wikipedia. Thesis or dissertation. URL: http://en.wikipedia.org/wiki/Thesis_or_dissertation, last checked on 2010-01-07.

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This code of conduct is a tool that helps students to be transparent about the use of GenAI and fits within the university's principles on academic integrity.

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- **Copyrighted materials:** For **lawfully obtained copyrighted material** you should preferably use M**365 Copilot Chat with your KU Leuven**

account. If you use another GenAI tool, you must be **absolutely certain** it does n**ot store or reuse the data you enter.**

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  Exam Regulations Article 84: *"Every conduct individual students display with which they (partially) inhibit or attempt to inhibit a correct judgement of their own knowledge, understanding and/or skills or those of other students, is considered an irregularity which may result in a suitable penalty. A special type of irregularity is plagiarism, i.e. copying the work (ideas, texts, structures, designs, images, plans, codes , ...) of others or prior personal work in an exact or slightly modified way without adequately acknowledging the sources. Every possession of prohibited resources during an examination (see article 65) is considered an irregularity."*

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- **Additional reading: KU Leuven guidelines on responsible use of Generative AI tools, and other information** (Dutch/English)

A few final words

If you are uncertain whether or not you should declare your use of GenAI tools, we suggest that you discuss this with your instructor or supervisor. It is always safer to declare GenAI use, even when it is not strictly required. However, declaring GenAI use does not entail that its use is

allowed; the right column in the table below provides more detailed instructions in this regard (code of conduct).

Moreover, advanced AI tools are evolving rapidly, and their capabilities have expanded significantly in a short period of time. As a result, we do not yet have all the answers about their responsible use. Finally, it is important to follow-up on the most recent evolutions in AI technologies, to have an open mind but also to be a bit cautious, to communicate with instructors, teaching assistants, supervisors and peers, to be as transparent as we can, and to learn together as we move along.

Student name: ..... **Student number:** .....

Please indicate with "X" whether it relates to a course assignment or to the Bachelor's or Master's thesis:

O This form is related to a **course assignment.**
Course name: ..... **Course number:** .....

This form is related to my O **Bachelor's** or O **Master's thesis.**
Title Bachelor's or Master's thesis: ..... **Supervisor:** .....

Daily supervisor: .....

Please indicate with "X":

O I **did not use** any GenAI assistance tool.

O I **did use** GenAI Assistance. In this case **specify which ones** (e.g. ChatGPT, M365 Copilot, ...): .....

GenAI assistance used as/for:	**Name of the GenAI tool(s) used** If helpful, also describe in which way you were using GenAI related to what is specified as code of conduct.	**Code of conduct:** **For each of the categories below, always take into account the important guidelines and remarks mentioned above (e.g. copyrighted data, sensitive or personal data,. . .).**
As a language assistant for reviewing or improving texts I wrote myself	*add text here*	This use is similar to using spelling and grammar check tools. In general, you do not have to refer to such kind of GenAI use in the text. **However, be careful:** • When using GenAI tools on texts you did not write yourself to improve the text, **you have to refer to the original source or author**, otherwise you are committing plagiarism and thus an irregularity.
As a paraphrasing tool	*add text here*	This use is allowed except when it is prohibited by the teacher or the program of study. You may paraphrase your own text or texts by an author other than yourself and take inspiration from what a GenAI tool or another tool suggests (unless it is not allowed). In general, you do not have to refer to such kind of GenAI use or other paraphrasing tools in the text. **However, be careful**: • If it entails text by an author other than yourself, **you are not allowed to include that paraphrased text without reference to the original source or author**. Without such reference, you would be committing plagiarism and thus an irregularity.

For translation aid to improve texts I wrote myself or to better understand text from others	*add text here*	This use is allowed except when it is prohibited by the teacher or the program of study. It is similar to using translation tools (Google translate, DeepL, ...). In general, you do not have to refer to such kind of GenAI use in the text. **However, be careful:** • **You are not allowed to include that translated text without reference to the original source or author.** Without such reference, you would be committing plagiarism and thus an irregularity. • Always check the translated text for correctness and meaning.
As a search engine to get information on a topic or to search for existing research on the topic	*add text here*	This use is similar to e.g. a Google search or checking Wikipedia. If you write your own text based on this information, you do not have to refer to the use of GenAI in the text. **You only have to refer to the existing research and references you have checked and used** (without such references you would be committing plagiarism and thus committing an irregularity). **However, be careful:** • Be aware that the output of the GenAI tool cannot be guaranteed as a 100% reliable source of information. The output may not be entirely correct and/or be limited due to the databases it uses. Moreover, knowledge evolves and may change over time; therefore, the database of the GenAI tool may not be up to date. Therefore, **verify the information and do not just copy-paste it** as you should understand and critically process everything you are writing.

For literature search	*add text here*	This use is comparable to e.g. a Google Scholar search. You do not have to refer to such kind of GenAI use; **you only have to refer to the literature references you have checked and used** (without such references you would be committing plagiarism and thus committing an irregularity). **However, be careful:** • Be aware that the search output is restricted to the database the GenAI tool is built on. After this initial search, look for scientific sources and conduct your own analysis of the source documents. **Interpret, analyse and process the information you obtained; verify it and do not just copy-paste it** as you should understand and critically process everything you are writing. • Be aware that some GenAI tools may **output no or wrong references.** As a student you are responsible for further checking and verifying the absence or correctness of references; do not just copy-paste it.
For generating programming code	*add text here*	Use of GenAI for coding is allowed except when it is prohibited by the teacher or the program of study. If used for coding, correctly mention the use of GenAI assistance in accordance with the instructions on the page on being transparant about the use of GenAI.

For generating new (research) ideas	*add text here*	This use of GenAI is allowed except when it is prohibited by the teacher or the program of study. Correctly mention the use of GenAI assistance in accordance with the instructions on the page on being transparent about the use of GenAI. **Be careful:** • Further verify in this case whether the idea is novel or not. It is likely that it is related to existing work. **If so, that existing work should be correctly referenced in the text** (without such reference you would be committing plagiarism and thus committing an irregularity).
For generating synthetic data	*add text here*	Use of GenAI for generating synthetic data is allowed, **provided that it is methodologically and ethically justifiable**, except when it is prohibited by the teacher or the program of study. Always correctly mention the use of GenAI assistance in accordance with the instructions on the page on being transparent about the use of GenAI. **Be careful:** • Always **carefully evaluate** the generated synthetic data for quality and possible bias since the output is highly dependent on the quality of the data on which the models are trained.

For generating blocks of text (other than the allowed use without referencing mentioned above)		According to Article 84 of the Regulations on Education and Examinations your text should allow to correctly and properly assess your own knowledge, understanding and skills. **Therefore, inserting blocks of text without quotes and a reference to GenAI assistance in your work is not allowed.** **Be careful:** • If it is really needed to insert a block of text from a GenAI tool, for instance because of the nature of your assignment, mention it as a citation by using quotes and correctly mention the use of GenAI assistance in accordance with the instructions page on being transparent about the use of GenAI. • However, in general, such GenAI use should be kept to an **absolute minimum**; you should always check the original sources.
For generating visuals, video or audio	*add text here*	Use of GenAI for generating visuals, audio or video is allowed except when it is prohibited by the teacher or the program of study. **Be careful:** • If used, refer to GenAI for visuals according to the usual referencing style, following the instructions on the page on referencing GenAI. • If you work with existing visuals, audio or video, **that existing work should be correctly referenced in the text** (without such reference you would be committing plagiarism and thus an irregularity). • Explain the usage in the methods section (if there is one) and optionally attach (or link to) the prompts with the full output (or history).

Other use (specify here; this may also include a combination of types of use mentioned above):	*add text here*	To make sure other use of GenAI is allowed within the course or thesis, and if so, the conditions that may apply, contact the teaching staff of the course or the supervisor of the thesis **beforehand** and explain the intended GenAI use. Also inform the programme director. Motivate how you would comply with Article 84 of the exam regulations. Explain the use and the added value of the AI tool you consider to use and how it is in accordance with the assignment or thesis and the KU Leuven guidelines on responsible use of Generative AI tools. Depending on the kind of GenAI use, it may be needed to properly reference it in the text, in accordance with the instructions page on being transparent about the use of GenAI.